

A lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ ternary system

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ABSTRACT

We developed a lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ ternary ceramic system comprising of the mole ratio as $\text{Li}_4\text{SiO}_4\text{:Li}_3\text{PO}_4\text{:LiBO}_2 = 60\text{:}40\text{:}20$ (termed as 40LP + 20B), exhibiting the ion conductivity of 5.7×10^{-6} S/cm at room temperature. From the X-ray diffraction patterns, the ternary solid solution has the orthorhombic phase structure with the increased unit cell volume due to the substitution of boron in lattice. The SEM images show that the high densification of the 40LP + 20B samples sintered at 900 °C for 6 h, and EDS mappings illustrate that the incorporating boron is distributed homogeneously in the grains and no precipitated impurity was observed on grain surface or around grain boundaries. The data of electrochemical impedance spectroscopy (EIS) reveal the low resistance from grain boundary, giving rise to the high bulk ion conductivity. The lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ system is of great potential as practical solid electrolyte for all-solid-state battery.

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1. Introduction

It is of great challenge to find new electrolytes suitable for the use in lithium or lithium ion batteries [1]. Stable solid electrolytes with high lithium ion conductivity are expected to offer several advantages over the commercialized liquid electrolytes such as higher thermal stability, absence of leakage and pollution, and a large electrochemical stability window [2]. To date, solid lithium ion conductors are extensively investigated in the materials of crystalline, glassy and composites with wide range of chemical compositions and different crystal structures, typically, $\text{Li}_3\text{xLn}_{2/3-\text{x}}\text{TiO}_3$ ($\text{Ln} = \text{La, Pr, Nd, Sm}$) with perovskite structure [3], $\text{Li}_1+\text{xTi}_2-\text{xM}_\text{x}(\text{PO}_4)_3$ ($\text{M} = \text{Al, Sc, Y, La}$) with NASICON structure [4], glass ceramics ($\text{Li}_7\text{P}_3\text{S}_{11}$), composites ($\text{Li}_2\text{S-SiS}_2\text{-Li}_3\text{PO}_4$) [5], and notable $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ with orthorhombic structure [6]. All these materials exhibit ion conductivities of the order of $10^{-4}\text{--}10^{-2}$ S/cm, but those compounds are not favorable as an electrolyte material for all-solid-state batteries due to their low stability in direct contact with commonly used intercalation electrode materials for battery applications. Garnet-related compounds, especially cubic- $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ and $\text{Li}_6\text{BaLa}_2\text{Ta}_2\text{O}_{12}$ attract more attention due to their high lithium ion conductivity ($\sim 10^{-4}$ S/cm) [7,8]. Regards of their chemical stability against electrode materials, the dense garnet ceramics are considered as solid state electrolytes for all-solid-state batteries. However, it is difficult to synthesize the pure garnet materials as well as to obtain dense ceramics with cubic polymorph for high ionic conductivity [9]. In this communication, we address a lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$

ternary system for its reasonable ion conductivity and good stability as well as easy fabrication, potential as practical solid electrolyte used in all-solid-state battery.

The binary system of $\text{Li}_3\text{PO}_4\text{-Li}_4\text{SiO}_4$ solid solution has been investigated extensively [10–14]. Li_4SiO_4 material has the structure generally described in terms of isolated $(\text{SiO}_4)^{4-}$ anionic clusters with ordered distribution of Li^+ ions over available cationic sites which are coordinated with 4, 5 or 6 oxygen atoms in a monoclinic unit cell ($P2_1/m$ space group). LiO_4 , LiO_5 and LiO_6 polyhedra are linked by edge- and corner-sharing. $\gamma\text{-Li}_3\text{PO}_4$ has a very similar orthorhombic structure ($Pmn2_1$ space group) formed of isolated $(\text{PO}_4)^{3-}$ anionic clusters with lithium ions in tetrahedral coordination [10]. These two lithium-containing compounds have ordered structure concerning Li ions, resulting in very low ion conductivity. Fortunately, mixing of these two compounds can dissolve each other into solid solution up to 60 mol%, leading to the vacancies or interstitial sites in the lattice for the increase of conductivity [11]. The solid solution of $\text{Li}_4-\text{xSi}_1-\text{xP}_\text{x}\text{O}_4$ with the composition of $0 < \text{x} < 0.12$ is stable in the Li_4SiO_4 monoclinic structure, while the $\text{Li}_4-\text{xSi}_1-\text{xP}_\text{x}\text{O}_4$ with composition $0.12 < \text{x} < 0.6$ is metastable in the orthorhombic structure similar to $\gamma\text{-Li}_3\text{PO}_4$ [12]. The ion conductivities of the solid solutions are several orders of magnitude greater than those of either end group. For example, Li ion conductivity of the samples with the compositions of $\text{x} \sim 0.5$ to 0.6 can achieve at the level of 10^{-6} S/cm and the activation energies is about $0.51\text{--}0.52$ eV¹³, while the ion conductivity for $\text{x} = 0.4$ is reported lower than that of the above compositions ($\sim 10^{-7}$ S/cm) [14]. It was noted that the doping amount of LiBO_2 from 0.5 to 5 wt.% as sintering agent into $\text{Li}_4-\text{xSi}_1-\text{xP}_\text{x}\text{O}_4$ ($\text{x} = 0.6$) solid solution could improve the ion conductivity due to the densification of ceramics [15]. Thus, it

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is intriguing that the ternary system of Li_4SiO_4 - Li_3PO_4 - LiBO_2 ceramic could be novel lithium ion conductor. In this case, up to the amount of 20 mol% LiBO_2 incorporated into the $\text{Li}_{4-x}\text{Si}_{1-x}\text{P}_x\text{O}_4$ system ($x = 0.4$) were investigated for the increase of ion conductivity. The effects of incorporating boron on phase structure and grain boundary as well as the densification are also discussed.

2. Experimental

The ceramic pellet samples were prepared by solid state reaction and sintered for densification. Li_4SiO_4 (Aldrich, 99%), Li_3PO_4 (Aldrich, 99%), B_2O_3 (Aldrich, 99%) and $\text{LiOH} \cdot \text{H}_2\text{O}$ (Aldrich, 99.9%) were used as the starting materials. The powders of the starting materials were mixed and formed into pellets by cold press under the pressure of 15–20 MPa for 2 min. The sample termed as 40LP was prepared with 60 mol% of Li_4SiO_4 and 40 mol% of Li_3PO_4 for $\text{Li}_{4-x}\text{Si}_{1-x}\text{P}_x\text{O}_4$ ($x = 0.4$); The sample termed as 40LP + 20B was prepared by mixing B_2O_3 , LiOH and 40LP sample according to the mole ratio of $\text{Li}_4\text{SiO}_4:\text{Li}_3\text{PO}_4:\text{LiBO}_2 = 60:40:20$. Regards of the solid state reaction and evaporating loss of lithium oxide at high temperature, the additional 10 mol% amount of LiOH was added for compensation of Li ion concentration in ternary system. To obtain homogenous solid solution, the mixture of starting materials were subjected to the solid state reaction at 900 °C and 2 h aging in the fluxing argon atmosphere for several times. For dense ceramic pellets, 40LP samples were sintered at 900 °C in fluxing argon atmosphere for 4 h, and the 40LP + 20B samples were sintered at 900 °C in fluxing argon atmosphere for 4 h and 6 h, respectively. The evaluated rate of increasing temperature was 10 °C/min, and cooling down naturally. The as-received ceramic pellets were termed as 40LP/4 h, 40LP + 20B/4 h and 40LP + 20B/6 h, respectively. Pellet dimensions and weights were measured before and after sintering. The relative density of as-received ceramic pellets was measured by the routine method based on Archimedes principle.

The phase structures were identified by X-ray diffraction method (XRD, Rigaku Ultima IV) using $\text{CuK}\alpha$ radiation ($\lambda = 1.54178 \text{ \AA}$). Scanning electron microscopy (SEM) was conducted of as received pellets using Zeiss ultra 55 SEM to observe the grain morphology from the fresh fracture surface of sample. Energy dispersive X-ray spectroscopy (EDS) was done to map the spatial distribution of boron in the grain or grain boundary by means of Oxford INCA penta FETx3 detector. The

electrical performance was examined by electrochemical impedance spectroscopy (EIS) using modulab ECS Solartron Analytical System. For this purpose, the pellets were polished to make their surface smooth and Au thick films were deposited by evaporation method as the blocking electrode. Impedance was recorded in frequency range 0.01 Hz to 1 MHz. An AC perturbation was applied with amplitude (rms) 10 mV. Complex impedance spectra were analyzed to reveal the resistances of grain and grain boundary as well as their associated capacitances.

3. Results and discussion

In Fig. 1, XRD pattern data are shown for entire range of solid solutions along with standard data for Li_4SiO_4 (PDF No. 37-1472) and γ - Li_3PO_4 (PDF No. 15-0760). From Fig. 1c, it is clear that the reflection peaks shift left from the original pattern of γ - Li_3PO_4 , typically marked in the planes (020), (021) and (002) respectively. It indicates that the as-received sample is the typical single solid solution phase of $\text{Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4$ with the orthorhombic structure similar to γ - Li_3PO_4 [10], and both the b-axis (010) and c-axis (001) are expanded dramatically from the original lattice structure while a-axis (100) is changed little. In Fig. 1d, it is observed that there are two set patterns, one representing the solid solution phase with orthorhombic structure, another representing the impurity ($\blacklozenge$) indexed as LiPO_3 according to PDF No.26-1177. Nevertheless, in Fig. 1e, the reflection peaks of as LiPO_3 impurity are almost disappeared, the dominate peaks are represented for the single solid solution phase. The occurrence of LiPO_3 impurity in the 40LP + 20B/4 h sample is responsible from the incorporation of boron into $\text{Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4$, and a little phosphate is precipitated from ternary solid solution system. In the ref. [15], the impurity of Li_2SiO_3 appeared in the additive of 5 wt.% LiBO_2 into $\text{Li}_{3.4}\text{Si}_{0.4}\text{P}_{0.6}\text{O}_4$ sample, resulted from extraction of lithia from the Li_4SiO_4 by reaction with LiBO_2 . However, regards of the longer sintering time from 4 h to 6 h, the LiPO_3 impurity was dissolved again for the complete ternary solid solution. It is also obvious that the reflection peaks of the 40LP + 20B sample sintering for 4 h and 6 h are left shifting from those of the original 40LP sample, specifically the peak indexed as (002) reflection shifting dramatically. It means that the lattice constant of solid solution is dilated due to the incorporation of boron to form the ternary solid solution system. It was assumed that increase of lattice constant upon

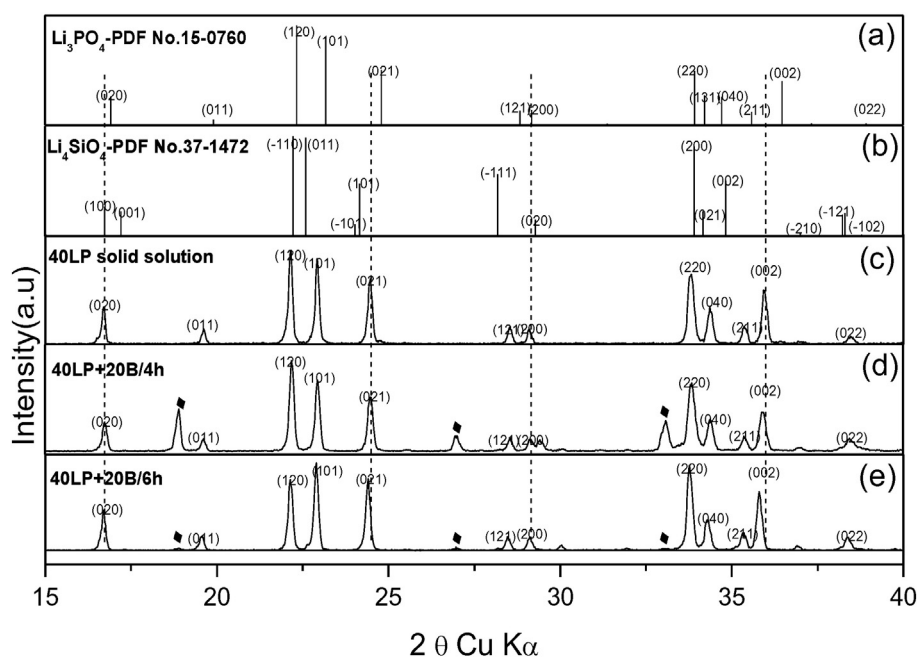


Fig. 1. XRD patterns corresponding to the standard data for Li_3PO_4 (a) and Li_4SiO_4 (b), 40LP/4 h (c), 40LP + 20B/4 h (d), and 40LP + 20B/6 h (e), respectively.

incorporating of boron leads to an expansion of “transport volume” for lithium ion and thus decreasing the activation energy of ion transport, which is benefit for the increased of lithium ion conductivity [16].

The microstructures and grain morphologies of the as-received pellets are observed by SEM, shown in Fig. 2. The morphologies of as-received 40LP, 40LP + 20B/4 h and 40LP + 20B/6 h are different from each other. The low densification and substantial anisotropic grain growth can be observed in the 40LP sample sintered without LiBO_2 , shown in Fig. 2a. In Fig. 2c, it is clear that 40LP + 20B/6 h sample has the dense microstructure and comparatively large grain size as well as the clear grain facet and boundary. In contrast, 40LP + 20B/4 h sample presents the blur grain boundary and the precipitation (white arrows pointed in Fig. 2b) on the grain surface represented for LiPO_3 impurity. The results indicate that the presence of LiBO_2 could help the grain growth and densification, and the longer sintering time would be of benefit for preventing from the precipitation and forming the complete solid solution. The element analysis was conducted by EDS in Fig. 2e and f. The atomic ratios were detected in 40LP and 40LP + 20B samples, which are consistent with the mole ratios of the starting materials.

The mapping of boron was also conducted in 40LP + 20B/6 h sample. In Fig. 2d, the red points represent the homogenous distribution of boron element on the equiaxial grains, and no precipitation was observed on grain surface or around grain boundary, indicating the boron is incorporated in the lattice of solid solution.

Ion conductivity was measured at room temperature by electrochemical impedance spectroscopy (EIS) with Au film as blocking electrode. Nyquist plots of all samples are presented in Fig. 3, showing low frequency range spike line followed by high frequency range depressed semicircle. The center of these semicircles are located below the real axis (Z'), indicating a non-Debye type of relaxation process and the presence of relaxation time distribution instead of a single relaxation time. The records of ion conductivity for the as-received samples are listed in Table 1. Total impedance was counted at the intercept of the arc with the real axis of Z' . The 40LP/4 h sample has the ion conductivity of 8.4×10^{-7} s/cm, which is consistent with the results from the ref. [14]. Nevertheless, the ion conductivity is approaching to 3.2×10^{-6} s/cm for 40LP + 20B/4 h and up to 5.7×10^{-6} s/cm for 40LP + 20B/6 h, respectively. It is noted that this ion conductivity of

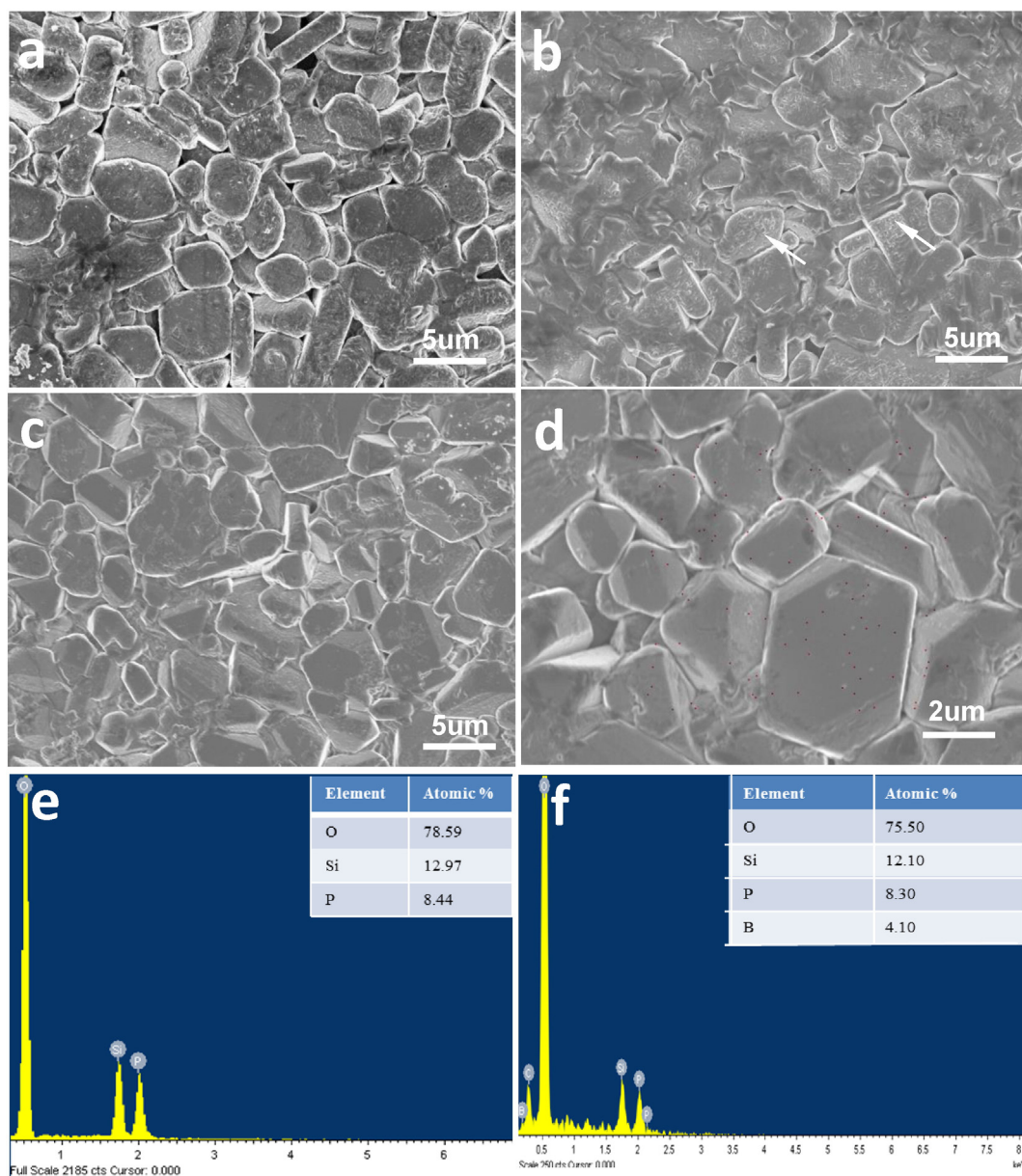


Fig. 2. SEM images of as-received samples sintered at 900 °C (a) 40LP/4 h, (b) 40LP + 20B/4 h, (c) 40LP + 20B/6 h, respectively; (d) Grain morphology of 40LP + 20B/6 h and the corresponding mapping of boron. EDS for the element analysis of 40LP (e) and 40LP + 20B/6 h (f).

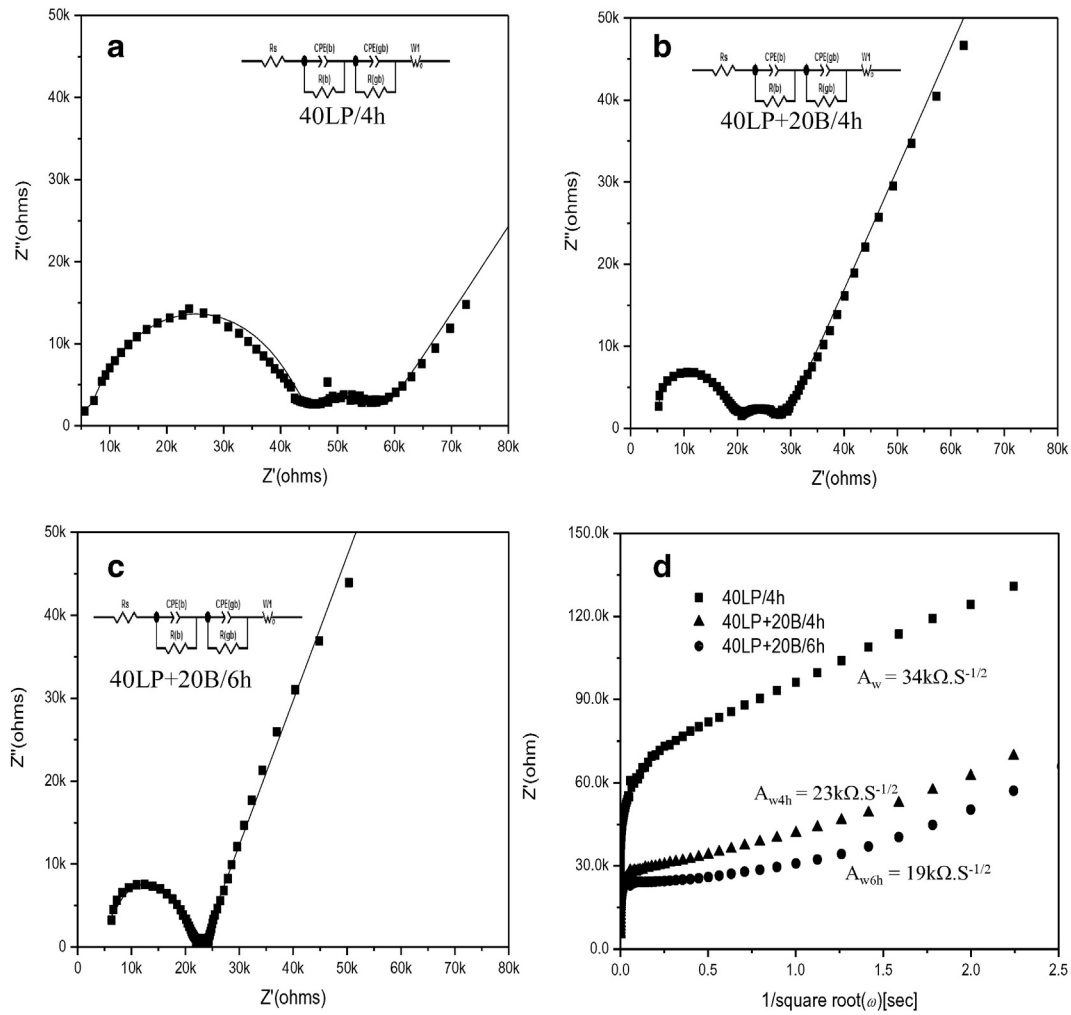


Fig. 3. Nyquist plots and the corresponding fitting by equivalent circuit model for the samples 40LP/4 h (a), 40LP + 20B/4 h (b), and 40LP + 20B/6 h (c), respectively. (d) Warburg plots for the above samples.

lithium ion conductor is higher than that of $\text{Li}_4 - x\text{Si}_1 - x\text{P}_x\text{O}_4$ solid solution ($x = 0.6$) with 0.5% LiBO_2^{15} and typical thin film electrolyte of LiPON [17]. The increase of ion conductivity can be assumed from the incorporation of boron into the lattice of the solid solution, resulting in the increased vacancies benefit for Li ions diffusion. The improved densification in 40LP + 20B/6 h samples is also responsible for the increase of ion conductivity.

It is well known that the fitting to Nyquist plots using equivalent circuit model could reveal the behaviors of lithium ion transport or resistance contributions in the ceramic pellets, identified as R_b (high frequency part of depressed semicircle), R_{gb} (medium frequency part of depressed semicircle) and their associated capacitance. The fitting results are clearly from complex impedance graph shown in Fig. 3. Spike line at low frequency shows polarization due to blocking electrode and to represent it, W_o is used in the equivalent circuit model. At lower frequencies, the blocking electrodes prevent ions from crossing the interface: ions periodically accumulate at one side of the sample while becoming depleted at the other [18]. According to Maxwell-Wagner effect, this spike at low frequency value is assigned to dielectric polarization between the sample surface and the electrode and is thus extrinsic to the sample [19]. The contributions from the components of equivalent circuit model are obtained by fitting experimental data listed in Table 2. It is shown that the grain resistance and grain boundary resistance decreased dramatically in the 40LP + 20B samples. From the Table 2, the calculated bulk resistance and capacitance in 40LP + 20B/4 h are almost same as those in 40LP + 20B/6 h, but the grain boundary

resistance (10457Ω) as well as the grain boundary capacitance ($1.59 \times 10^{-6} \text{ F}$) in 40LP + 20B/4 h is much higher than those (3499Ω for resistance and $2.62 \times 10^{-9} \text{ F}$ for capacitance, respectively) of grain boundary in 40LP + 20B/6 h. These results are attributed from the precipitated impurity of LiPO_3 from the solid solution in 40LP + 20B/4 h, giving rise to the higher resistance for lithium ion transport, and exhibiting the lower ion conductivity than that of 40LP + 20B/6 h.

Moreover, the mobility of Li ions at low frequency was investigated by the determination of Warburg coefficient, which is associated to the diffusion of ions. The finite Warburg element Z_w is given as [20]

$$Z_w = A_w w^{-1/2} (1 - j) \tanh(\delta(jw/D)^{1/2})$$

With δ = Nernst diffusion layer thickness, w = angular frequency, D = diffusion coefficient of electro active species i.e. Li ions, $j = (-1)^{1/2}$ and A_w = Warburg coefficient. With a high numerical value of $\delta/D^{1/2}$, the hyperbolic tangent term tends towards unity and the impedance plot shows a straight line instead of semicircle at low frequency values. Thus the small value of D will favor the semi-infinite diffusion i.e. inclined straight line tail which is also the case shown in Fig. 3a, b, and c. Warburg coefficient is known to follow a dependence on D and C of the form

$$A_w = RT/2^{1/2} n^2 F^2 * 1/CD^{1/2}$$

Where R , T , F and n stand for the usual electrochemical parameters. C is the concentration of diffusing species which remains roughly constant when the maximum concentration gradient is reached and then A_w becomes an exclusive function of D . Thus it indicates that the higher A_w value represent the lower mobility of the diffusion process [20]. The

Table 1
Measured ion conductivity of as-received samples.

Sample type	Diameter D (cm)	Thickness T (cm)	Resistance R (kΩ)	Ion conductivity σ (S/cm)
4LP/4 h	1.24	0.057	57	8.4×10^{-7}
4LP + 20B/4 h	1.24	0.09	23	3.2×10^{-6}
4LP + 20B/6 h	1.24	0.125	18	5.7×10^{-6}

Note: Ion conductivity is calculated as the formula $\sigma = 1/R * (4 t/\pi D^2)$.

value of A_w was obtained by the slope of the Warburg plot, which is a linear plot of the real impedance (R) against the reciprocal of the square root of the frequency (w), shown in Fig. 3d. It is clear that the value for A_w in the case of 40LP/4 h samples is much higher than those of 40LP + 20B/4 h and 6 h samples. The low A_w value of 40LP + 20B/6 h sample without the precipitated LiPO_3 performs the higher mobility of the Li ion diffusion, beneficial to the good interface to the electrode for practical use in all-solid state battery.

The developed lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ ternary system is free of transition metal, possible for the sufficient stability against lithium. However, it seems hard to be the electrolyte applied for solid state battery due to the relative lower ion conductivity (5.7×10^{-6} S/cm). It is suggested in next investigation to deal with the doping of super-valent cations substituting lithium ions into this ternary solid solution for the enhancement of the ion conductivity.

4. Conclusion

As a summary, we developed a lithium ion conductor in $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ ternary system, which is comprised of the composition as the mole ratio of $\text{Li}_4\text{SiO}_4\text{:Li}_3\text{PO}_4\text{:LiBO}_2 = 60\text{:}40\text{:}20$, exhibiting the ion conductivity of 5.7×10^{-6} S/cm and Warburg coefficient (A_w) $17 \text{ k}\Omega \cdot \text{s}^{-1/2}$ at room temperature. The X-ray diffraction patterns reveal that the ternary solid solution has the orthorhombic phase structure with the increased unit cell volume due to the incorporation of boron in lattice. The high densification was observed in the above samples sintered at 900 °C for 6 h by means of SEM. The element analysis illustrated that the boron is distributed homogenously in the grains and no impurity was observed around grain boundaries. The fitting to the experimental Nyquist plots reveal that the developed lithium ion conductor exhibits the low grain resistance and grain boundary resistance as well as the low capacitance for grain boundaries. It is still under searching for higher ion conductivity in this ternary system of $\text{Li}_4\text{SiO}_4\text{-Li}_3\text{PO}_4\text{-LiBO}_2$ as ceramic electrolyte for practical applications.

Table 2
Simulated equivalent circuits for Nyquist plots of samples.

Circuit element	40LP(4 h)	40LP + 20B/4 h	40LP + 20B/6 h
Rs(ohm)	6228	3798	6028
CPE(b)-T(F)	6.49E-11	0.1026E-11	1.644E-11
CPE(b)-P	0.79	0.95	0.9
Rb(ohm)	28,998	13,823	13,118
CPE(gb)-T(F)	2.4E-6	1.59E-6	2.620E-9
CPE(gb)-P	0.39	0.48	0.99
Rgb(ohm)	23,170	10,457	3499
W-R(ohm·cm)	0.1942	2398	880.5
W-T(sec)	7.36E-7	0.025–9	9.3E-3
W-P(ohm)	0.43	0.31	0.33

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References

- [1] J.B. Goodenough, K.-S. Park, J. Am. Chem. Soc. 135 (2013) 1167–1176.
- [2] P. Knauth, Solid State Ionics 180 (2009) 911–916.
- [3] M. Itoh, Y. Inaguma, W. Jung, L. Chen, T. Nakamura, Solid State Ionics 70–71 (1994) 203–207.
- [4] H. Aono, E. Sugimoto, Y. Sadaoka, N. Imanaka, G. Adachi, J. Electrochem. Soc. 136 (1989) 590–591.
- [5] T. Minami, A. Hayashi, M. Tatsumisago, Solid State Ionics 177 (2006) 2715–2720.
- [6] N. Kamaya, K. Homma, Y. Yamakawa, M. Hirayama, R. Kanno, M. Yonemura, T. Kamiyama, A. Mitsui, Nat. Mater. 10 (2011) 682–686.
- [7] R. Murugan, V. Thangadurai, W. Weppner, Angew. Chem. Int. Ed. 46 (2007) 7778–7781.
- [8] V. Thangadurai, H. Kaack, W. Weppner, J. Am. Ceram. Soc. 86 (2003) 437–440.
- [9] E. Rangasamy, J. Wolfenstine, J. Sakamoto, Solid State Ionics 206 (2012) 28–32.
- [10] M. Smiati, D. Petit, F. Gourbilleau, F. Chaput, J.P. Boilot, Solid State Ionics 48 (1991) 213–223.
- [11] T. Asai, S. Kawai, Solid State Commun. 39 (1980) 891–893.
- [12] A. Khorassani, G. Izquierdo, A.R. West, Mater. Res. Bull. 16 (1981) 1561–1567.
- [13] Y.W. Hu, I.D. Raistrick, R.A. Huggins, J. Electrochem. Soc. 124 (1977) 1240–1242.
- [14] Y.W. Hu, I.D. Raistrick, R.A. Huggins, Mater. Res. Bull. 11 (1976) 1227–1230.
- [15] L. Zhang, L. Cheng, J. Cabana, G. Chen, M. Döeff, T.J. Richardson, Solid State Ionics 231 (2013) 109–215.
- [16] E.A. Secco, Solid State Commun. 66 (1988) 921–923.
- [17] B. Fleutot, B. Pecquenard, H. Martinez, M. Letellier, A. Levasseur, Solid State Ionics 186 (2011) 29–36.
- [18] J. Jamnik, J. Maier, Phys. Chem. Chem. Phys. 3 (2001) 1668–1678.
- [19] J.H. Hwang, K.S. Kirkpatrick, T.O. Mason, E.J. Garboczi, Solid State Ionics 98 (1997) 93–104.
- [20] J.R. Scully, D.C. Silverman, M.V. Kending (Eds.), Electrochemical Impedance Analysis and Interpretation, ASTM (publication code number (PCN) 04-011880-27), 1993.